

ECN 712: Microeconomic Analysis I

EE SCHLEE FALL 2025

Office: CRTVC 490D; email: ed.schlee@asu.edu

Hours: After class (in person); virtual or in person by appt.

TA: Titus Gao, <ygao190@asu.edu>.

We'll begin a study of the core of microeconomic theory, including firm and consumer theory and their interaction in markets and games. The prerequisites are a *thorough* understanding of elementary microeconomics, calculus, linear algebra, and above all simple logic.

Required Texts: *Microeconomic Theory*, by A. Mas-Colell, M. Whinston and J. Green (**MWG**); D. Kreps, *Microeconomic Foundations I* (**K**).

Useful References. (A (★) means it is available free online—see the Canvas page for links.)

GENERAL ECONOMIC THEORY: G. Jehle and P. Reny, *Advanced Microeconomic Theory* 2e (2001); D. Kreps, *Microeconomic Foundations II*.

MATHEMATICS: R. Sundaram, *A First Course in Optimization* (1996); D. Topkis, *Supermodularity and Complementarity* (1998); P. Halmos, *Naive Set Theory* (1960).

GAME THEORY: R. Gibbons, *Game Theory for Applied Economists* (1992); M. Osborne and A. Rubinstein, *A Course in Game Theory*★ (1994); D. Fudenberg and J. Tirole, *Game Theory* (1991).

SPECIAL TOPICS: J. Tirole, *The Theory of Industrial Organization* (1988); X. Vives, *Oligopoly Pricing* (1999); J. Hirshleifer and J. Riley, *The Analytics of Uncertainty and Information* (1992); C. Gollier, *The Economics of Uncertainty* (2001).

ELEMENTARY MICROECONOMICS: H. Varian, *Intermediate Microeconomics*, any edition.

ESSENTIAL CLASSICS: P. Samuelson, *Foundations of Economic Analysis* (rev. 1984); G. Debreu, *Theory of Value*★ (1959 (from now on **TV**)).

Grading: I base your course grade on two equally-weighted exams, the first covering Part I of the outline and the second Part II. I also assign weekly problem sets, to be graded coarsely on a ✓+, ✓, ✓− basis. I only use these scores for your course grade in borderline cases. I encourage you to discuss the problem sets with others, but do so judiciously: If you rely too much on hints from your colleagues, then all you'll learn is how to solve the problems on that assignment, not how to *think economically*. I cannot overemphasize the importance of carefully working through and thinking about the problem sets. I expect that the first exam will be the first week of October. The second exam is scheduled for the final exam period, **Thursday, 11 December, 12:10 – 2:00 PM**

Discussion Section: Titus will meet with you Tuesdays at 6:05 PM in the usual classroom to go over the weekly problem sets. (No meeting on the first Tuesday.)

Course Outline

PART I: CONSUMERS, PRODUCERS, COMPETITIVE MARKETS, AND COMPARATIVE STATICS

- Consumer Theory (3 weeks; MWG, chapters [chs] 1-4; Kreps, chs 1-4,10-13; TV chs 2,4)
 - ‡ Budgets, Preferences, and Utility
 - ‡ Demand; Revealed Preference; Aggregation; Consumer's Surplus
- Producer Theory (3 weeks; MWG, ch 5; Kreps, chs 9, 13; TV, ch 3)
 - ‡ Technology
 - ‡ Profit Maximization; Cost Minimization
 - ‡ Monotone Comparative Statics
- Competitive Markets and Efficiency–Partial Equilibrium (1.5 weeks; MWG, ch 11; Kreps ch 13.2, TV chs 5-6.)
 - ‡ Competitive Equilibrium
 - ‡ The First Welfare Theorem

PART II: DECISIONS AND GAMES

- Decision Theory (2 weeks; MWG, ch 6; Kreps, chs 5-6; TV, ch 7)
 - ‡ Expected Utility; Risk Aversion; Stochastic Dominance
 - ‡ Portfolio Choice and Insurance Demand
 - ‡ Subjective Probability
- Monopoly (1 week; MWG, chapter 12)
- An Introduction to Game Theory; Oligopoly Theory (4 weeks; MWG, chapters 7-9, 12, 13B.)
 - ‡ Strategic and Extensive Form Games (of Complete Information); Dominance; Rationalizability; Nash Equilibrium; Subgame Perfection
 - ‡ Static Oligopoly Models: Cournot; Bertrand; Stackelberg
 - ‡ Supermodular Games
 - ‡ Repeated Games and Some Folk Theorems

Interpretation of Letter Grades

Economics 712 serves a varied audience: Ph.D. students in Economics and Finance, graduate students in other disciplines, and ambitious undergraduates. My grading policy reflects this diverse audience. I will assign each exam a numerical score and a letter grade. Economics Ph.D. students preparing for the qualifying exam should note the following interpretation of these letter grades.

A+: truly exceptional (rarely given)

A: outstanding

A-: excellent (would clearly pass the qualifying exam in microeconomics)

B+: good (should pass the qualifying exam)

B: fair (marginal performance on the qualifying exam)

B-: barely passing (unlikely to pass the qualifying exam)

C: failing

A Short Reading List

I. CONSUMERS, PRODUCERS AND COMPETITIVE MARKETS

• Consumer Theory

–Debreu, G. (1959), chs. 2 and 4.¹

–Samuelson, P. “A Note on the Pure Theory of Consumer’s Behavior,” *Economica* (February, 1938): 61-71.

–Richter, M.K. “Rational Choice,” in Chipman, J. S., D. McFadden and M. K. Richter. 1990. *Preferences, Uncertainty and Optimality* Westview Press.

–Gorman, W. M. “Community Preference Fields,” *Econometrica* 21 (1953): 63-80.

–McKenzie, L. “Consumer Theory without a Utility Index,” *Review of Economic Studies* 24 (1957): 185-89.

–Blundell, R., P. Pashardes and G. Weber, “What Do we Learn About Consumer Demand Patterns from Micro Data?,” *American Economic Review*, 83 (1993): 570-597.

–Manser, M. and R. McDonald, “An Analysis of Substitution Bias in Measuring Inflation,” *Econometrica* 56 (1988): 909-930.

–Hildenbrand, W. *Market Demand: Theory and Empirical Evidence*. Princeton University Press (1994).

–Diewart, W., M. Avriel, I. Zang, “Nine kinds of quasiconcavity and concavity,” *Journal of Economic Theory* 25 (3): 397-420 1981.

–Kreps, D., “A Representation Theorem for ‘Preference for Flexibility’,” *Econometrica* 47: 565-577 1979.

–Milgrom, P. and I. Segal, “Envelope Theorems for Arbitrary Choice Sets,” *Econometrica* 70 (2002): 583-602.

–Sinander, L. “The converse envelope theorem,” *Econometrica* (2022): 2795-2819.

–Khan, M.A. and E.E. Schlee, “Money metrics in applied welfare analysis: a saddlepoint rehabilitation,” *International Economic Review*, 63.1 (2022): 189-210.

–Jerison, M. ‘Social Welfare and the Unrepresentative Representative Consumer,’ (2023): 5-28. *Journal of Public Economic Theory*.

¹The reference “Debreu (1959), ch. n” refers to the chapter in his monograph, *The Theory of Value*

• Producer Theory

- Debreu (1959), ch. 3.
- Shepard, R, *The Theory of Cost and Production Functions*, 1970: Princeton.
- Duality McFadden, D., “Cost, Revenue, and Profit Functions,” in M. Fuss and D. McFadden (eds.) *Production Economics: A Dual Approach to Theory and Applications*.
- Solow, R, “Technical Change and the Aggregate Production Function,” *The Review of Economics and Statistics*, 39 (1957): 312-320.

• Monotone Comparative Statics

- Topkis, D., “Comparative Statics of the Firm,” *Journal of Economic Theory*, 67, (1995): 370-401.
- Milgrom, P and C. Shannon, “Monotone Comparative Statics,” *Econometrica*, 62 (1994): 157-180.
- Shannon, C, “Weak and Strong Comparative Statics,” *Economic Theory*, 1995.
- Quah, John K.-H. “The comparative statics of constrained optimization problems.” *Econometrica* 75.2 (2007): 401-431.”
- Quah, John K.-H. and B. Strulovici, “Comparative Statics, Informativeness, and the Interval Dominance Order,” *Econometrica*, 77 (2009).
- Quah, John K.-H and Pawel Dzielwski. “Comparative statics with linear objectives: normal demand, monotone marginal costs, and ranking multi-prior beliefs,” *Econometrica*, Vol. 92, No. 1 (2024), 167–200.
- Amir, Rabah, and David Rietzke. “A Comment on ‘Monotone Comparative Statics’.” *Econometrica* 93.4 (2025): 1481-1490.

• Competitive Markets

- Marshall, A. *Principles of Economics*, 8th ed. (1920), New York: Macmillan.
- Hayek, F.A. The Use of Knowledge in Society, *Am Econ Rev* (1945).
- Willig, R.D., “Consumer’s Surplus without Apology,” *American Economic Review*, 66 (1976): 589-597.
- Vives, X., “Small Income Effects: A Marshallian Theory of Consumer’s Surplus and Downward-Sloping Demand,” *Review of Economic Studies*, 54 (1987): 87-103.
- Hausman, J. “Exact Consumer’s Surplus and the Deadweight Loss,” *American Economic Review*, 71 (1981): 662-676.
- Debreu, G. “A Classical Tax-Subsidy Problem,” *Econometrica*, 21 (1954).
- Radner, R, “A Note on the Theory of Cost-Benefit Analysis in the Small,” in *Capital, Investment and Development* Basu, K. and M. Majumdar and T. Mitra (eds), (1993) Blackwell, pp. 129-141.
- Weitzman, M., “Prices vs. Quantities,” *The Review of Economic Studies*, 41, (1974): 477-491
- Akerlof, G., “The Market for Lemons: Qualitative Uncertainty and the Market Mechanism,” *Quarterly Journal of Economics* 84 (1970), 488-500.
- Schlee, Edward E. “The value of information in efficient risk-sharing arrangements.” *American Economic Review* 91.3 (2001): 509-524.

II. DECISIONS AND GAMES

• Decision Theory; Money Outcomes and Risk Aversion

- Debreu (1959), ch. 7.
- Savage, L.J. *Foundations of Statistics*, 1954.
- Kreps, D. *Notes on the Theory of Choice*, Underground Classics, 1988.
- Herstein, I. and J. Milnor, “An Axiomatic Approach to Measurable Utility,” *Econometrica* (1953) 21: 291-297.
- Machina, M., “‘Expected Utility’ Analysis without the Independence Axiom,” *Econometrica* 50 (1982): 277-323.
- Machina, M., “Choice Under Uncertainty: Problems Solved and Unsolved,” *Journal of Economic Perspectives*, 1, 1987, 121-154.
- F.P. Ramsey “Truth and Probability,” (1926) in Ramsey, 1931, *The Foundations of Mathematics and other Logical Essays* [available: <http://homepage.newschool.edu/het//texts/ramsey/ramsess.pdf>].
- Ellsberg, Daniel (1961), “Risk, Ambiguity, and the Savage Axioms,” *Quarterly Journal of Economics* 75 (4): 643–669.
- Anscombe, F. and R.J. Aumann, “A Definition of Subjective Probability,” *The Annals of Mathematical Statistics*, 34, (1963): 199-205.
- Arrow, K. *Aspects of the Theory of Risk- Bearing*, Helsinki, 1965.
- Pratt, J. (1964). “Risk Aversion in the Small and in the Large,” *Econometrica*, 32, 122-36.
- Athey, S., “Monotone Comparative Statics Under Uncertainty,” *Quarterly Journal of Economics*, CXVII (2002): 187-223.
- Borch, K., “A Note on Uncertainty and Indifference Curves,” *Review of Economic Studies*, 36, 1969, 1-4.
- McCloskey, D., “English Open Fields as Behavior Towards Risk,” *Research in Economic History* 1 (1976): 124-170.
- Arrow, K., “Uncertainty and the Welfare Economics of Health Care,” *American Economic Review* 53 (1963): 941- 973.
- Blackwell, D. “Equivalent comparisons of experiments” *Ann. Math. Statistics* 24, (1953): 265-272.

• Monopoly

- Harberger, A. “Monopoly and Resource Allocation,” *American Economic Review* 44 (1954): 77-87.
- Ramsey F.P., “A Contribution to the Theory of Taxation,” *Economic Journal*, 37 (1927): 47-61.
- Oi, W., “A Disneyland Dilemma: Two Part Tariffs for A Mickey Mouse Monopoly,” *Quarterly Journal of Economics* (1971).
- Arrow, K. “Economic Welfare and the Allocation of Resources for Invention” in the N.B.E.R. *The Rate and Direction of Inventive Activity*, ed. R. Nelson, (1962) Princeton University Press.
- Spence, A. Michael “Monopoly, Quality and Regulation,” *Bell Journal of Economics*, 6 (1975): 417-29.
- Coase, R. “Durability and Monopoly” in *Journal of Law and Economics*, 15, (1972):143-49, .

• Game Theory

- von Neumann, John. “Zur Theorie der Gesellschaftsspiele,” *Mathematische Annalen* 1928, 100, pp. 295- 320; translated by S. Bargmann as “On the Theory of Games of Strategy,” in *Contributions to the theory of games*. Vol. 4. Eds.: Albert Tucker and R. Duncan Luce. Princeton: Princeton U. Press, 1959, pp. 13-42.
- Von Neumann, J., and O. Morgenstern. *Theory of games and economic behavior*. (1944).
- Nash, J. “Non-Cooperative Games,” *Ann. Mathematics* 54 (1951): 286-95.
- Schelling, T. *The Strategy of Conflict*, Cambridge Press: 1960.
- Kuhn, H.W., “Extensive games and the problem of information,” in *Classics in Game Theory*.
- Selten, Reinhard. “Spieltheoretische behandlung eines oligopolmodells mit nachfragertrgheit: Teil i: Bestimmung des dynamischen preisgleichgewichts.” *Zeitschrift fur die gesamte Staatswissenschaft/Journal of Institutional and Theoretical Economics* H. 2 (1965): 301-324.
- Bernheim, D. “Rationalizable Strategic Behavior,” *Econometrica* 52 (1984): 1007-1028.
- Pearce, D. “Rationalizable Strategic Behavior and the Problem of Perfection,” *Econometrica* 52 (1984): 1029-1050.
- Milgrom, P. and J. Roberts, “Rationalizability, learning, and equilibrium in games with strategic complementarities,” *Econometrica*, 58 (1990): 1255-1275.
- Friedman, J. “A Noncooperative Equilibrium for Supergames,” *Review of Economic Studies* (1971).
- Abreu, D., “On the Theory of Infinitely Repeated Games with Discounting,” *Econometrica*, 56 (1988): 383-396.
- Benoit, J. and Krishna, V., “Finitely Repeated Games,” *Econometrica* 53 (1985): 905-22.
- Strotz, Robert Henry. “Myopia and inconsistency in dynamic utility maximization.” *The Review of Economic Studies* 23.3 (1955): 165-180.
- Kydland, F., and E. Prescott, “Rules Rather than Discretion: The Inconsistency of Optimal Plans”. *Journal of Political Economy* (1977): 473-492.
- Kochov, A, and Y Song. “Intertemporal hedging and trade in repeated games with recursive utility.” *Econometrica* 91 (2023): 2333-2369.

• Oligopoly Theory

- Cournot, A, *Researches into the Mathematical Principles of the Theory of Wealth*, 1838.
- Stigler, G. “A Theory of Oligopoly” *Journal of Political Economy* (1964).
- Selten, Reinhard, “The chain store paradox,” *Theory and Decision* 9 (1978): 127-159.
- Kreps, D. and Scheinkman, “Quantity Precommitment and Bertrand Competition Yield Cournot Outcomes,” *Bell Journal* (1983).
- Bernheim, D. and M. Whinston, “Multimarket Contact and Collusive Behavior” *RAND Journal of Economics* 21 (1990): 1-26.
- Dixit, A. “The Role of Investment in Entry-Deterrence,” *Economic Journal* 90 (1980): 95-106.
- Bulow, J., J. Geanakoplos and P. Klemperer “Multimarket Oligopoly: Strategic Substitutes and Complements” *J. Polit. Econ.*, 93 (1985): 488-511.
- Fudenberg, D. and J. Tirole, “The Fat-Cat Effect, the Puppy-Dog Ploy, and the Lean and Hungry Look,” *American Economic Review*, 74 (1984), 361-366.
- Sutton, J. *Sunk Costs and Market Structure*. MIT Press: 1991.
- Schlee, Edward E. “Surplus maximization and optimality.” *American Economic Review* 103.6 (2013): 2585-2611.
- Carvajal, A., R. Deb, J. Fenske, and J. Quah, “Revealed preference tests of the Cournot model,” *Econometrica*, 81 (2013): 2351-2379.